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A new temperature profiling probe for investigating groundwater-surface water interaction

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Key Points:

- Low-profile design allows for continuous monitoring in harsh environments
- Does not require removal or disruption for data retrieval
- Designed to use low-cost available self-logging temperature sensors that are easily replaced

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Abstract Measuring vertically nested temperatures at the streambed interface poses practical challenges that are addressed here with a new discrete subsurface temperature profiling probe. We describe a new temperature probe and its application for heat as a tracer investigations to demonstrate the probe's utility. Accuracy and response time of temperature measurements made at six discrete depths in the probe were analyzed in the laboratory using temperature bath experiments. We find the temperature probe to be an accurate and robust instrument that allows for easily installation and long-term monitoring in highly variable environments. Because the probe is inexpensive and versatile, it is useful for many environmental applications that require temperature data collection for periods of several months in environments that are difficult to access or require minimal disturbance.

1. Introduction and Motivation

Temperature measurements in stream environments are routinely made for researching ecological, hydrological, and many other processes that are important to the preservation of riverine resources [*Poole and Berman*, 2001]. In-stream temperature is a key parameter for monitoring the suitability of fisheries habitat, and streambed temperature data are useful for estimating surface water and groundwater exchange [*Brunke and Gonser*, 1997; *Briggs et al.*, 2013; *Constantz*, 2008; *Rau et al.*, 2014]. The usefulness of temperature data for environmental applications has created an important need for measuring devices that can withstand the harsh stream environment and provide accurate and robust data sets.

Streams are highly variable in space and time due to flow, geomorphologic, and hydrogeologic conditions. As such, temperature measurements made at many spatial locations and over long periods of time are needed to characterize riverine systems. Temperature measuring instruments can be deployed broadly in stream environments and thus provide a means of better understanding these systems [*Loheide and Gorelick*, 2006; *Selker et al.*, 2006]. However, relatively few in-situ temperature probes have been designed specifically for surface water systems. Rather, instruments designed for other applications [e.g., food transport] have been adapted for use in traditional monitoring approaches [e.g., *Shope et al.*, 2012; *Naranjo et al.*, 2012]. Although recent work has led to temperature probes designed specifically for streams, there still exists many practical problems including difficulty with probe installation, data collection and retrieval, and probe durability [*Wolaver and Sharp*, 2007].

Sediment temperature measured over long durations can be used in models to derive time-series seepage rates with dynamically changing surface flow conditions. Commonly, investigations have taken advantage of self-contained temperature sensors that collect and store continuous records. The use of thermochron iButtons (Maxim Integrated, DS1921H/DS1921Z High Resolution Thermochron iButton Devices, http://datasheets.maximintegrated.com/en/ds/DS1921H-DS1921Z.pdf, 2013) are attractive for use in monitoring temperature due to the costs ($30 USD), self-contained memory, small size (17 mm diameter) and with laboratory calibration, improved accuracy to ± 0.1°C [*Hester et al.*, 2009, *Naranjo et al.*, 2012]. Researchers have used iButtons for hydrological investigations [*Johnson et al.*, 2005; *Hubbart et al.*, 2005; *Hester et al.*, 2009; *Shope et al.*, 2012]; however, sensor failure occurs due to exposure to water [*Wolaver and Sharp*, 2007]. Waterproof enclosures (Maxim Integrated, model DS9107) made available by the manufacturer are available at a significant increase in cost to each sensor and diameter ($25 USD). Waterproofing the iButtons in silicone or plastic has been successful with no significant impact to accuracy or precision [*Lundquist and Lott*, 2008; *Roznik and Alford*, 2012]. Commercially available polyurethane sealed iButtons (PSB, Alpha Mach, Inc,

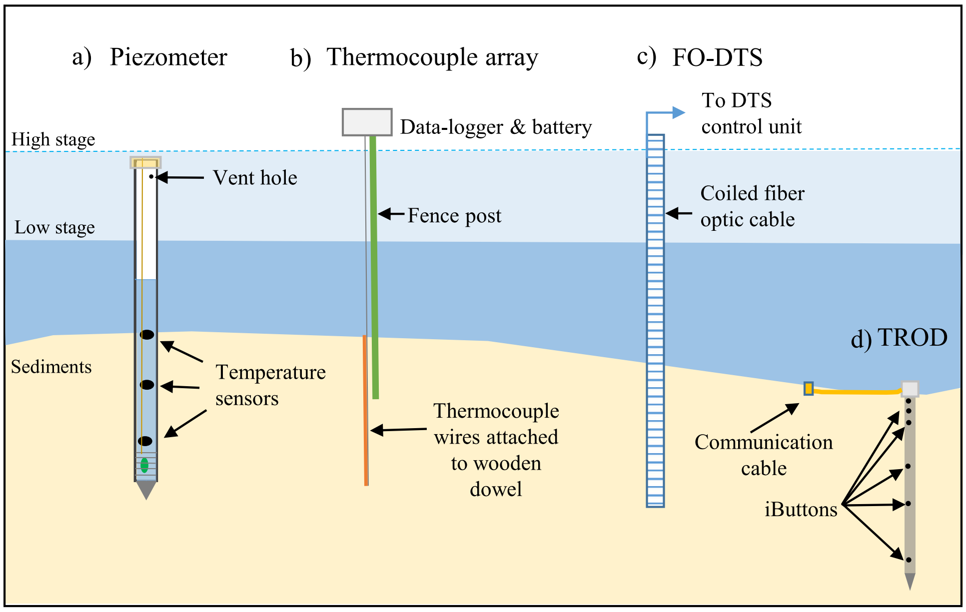

Figure 1. Monitoring sediment temperature in a surface water system with (a) piezometer, (b) thermocouple array, (c) fiber optic distributed temperature sensor (FO-DTS), and (d) the new subsurface temperature profiling probe (TROD). The dash line indicates high stage conditions with water submerging the top of the piezometer. Deployment of temperature sensors using Figures 1a–1c may be limited to shallower surface water depths and can be susceptible to damage at high stage from increased velocity or debris, whereas, the TROD is nonobstructive to flow thus allowing for installation in variety of environments. Data are obtained from the piezometer by removing temperature sensors from the piezometer and downloading data individually. Data are retrieved from the TROD through the submersible communication cable or by direct contact with the head of the probe using reader (shown in Figure 2b) for all sensors simultaneously.

model iBCod) have been used successfully in piezometers installed within a riffle-pool sequence [*Naranjo et al.*, 2012] and have been tested to depths of 900 m (Type Z iButton DS1921Z). The small size of the iButton on the PSB allows placement into smaller-diameter pipe that provide a reduced lag of the diurnal signal caused by thermal skin effects [*Cardenas*, 2010].

Discrete subsurface temperature measurements have been made in stream environments using three basic approaches (Figure 1): (1) Individual self-logging temperature sensors inside an in-stream pipe [e.g., *Constantz et al.*, 2002; *Rau et al.*, 2010; *Naranjo et al.*, 2012]; (2) Thermocouple or thermistor type arrays that measure temperature at multiple points and are wired into a single electronic data-logging device [e.g., *Constantz et al.*, 2001, 2002]; (3) Fiber-optic type distributed temperature sensor (FO-DTS) that measure temperature at many locations with depth [*Vogt et al.*, 2010; *Briggs et al.*, 2013]. Each of these approaches have pros and cons; the suitability of each of these approaches is dependent on the environmental application and specific needs of the study. Comparisons between the aforementioned methods have been reported extensively in the literature [e.g., *Constantz et al.*, 2002; *Lundquist and Lott*, 2008; *Gordon et al.*, 2013; *Rau et al.*, 2014]. Here we focus on improvements in the design of approach 1 and present a design consisting of multiple self-logging temperature sensors in a 1 m probe. This instrument is relatively easy to install, provides flexible data retrieval, and very little disturbance to the stream.

Previous studies have developed similar temperature probes for use in heat as tracer applications. For example, iButtons have been embedded into steel and wooden rods for used in estimating flux in restored streams [*Fanelli and Lautz*, 2008; *Gordon et al.*, 2013]. Temperature probes have also been designed with multiple sensors placed inside an enclosed PVC pipe [e.g., *Rau et al.*, 2010; *Tonina et al.*, 2014] or more commonly, suspended inside piezometers for both temperature and pressure observations (Figure 1a). These approaches are innovative in their implementation; however, they have practical limitations. Sensors placed in enclosed pipes can installed near flush with sediments limiting damage and visibilty [*Tonina et al.*, 2014]. However, since data are stored independently within each sensor, the entire array must be removed for data retrieval. To ensure data continuity, it is advantageous not to move the temperature sensors for retrieving data to avoid data spikes and potential shifts in sensor depth during replacement. Further, measuring temperature in piezometers that are exposed to flowing water can be susceptible to damage from floating debris and high velocities. In-stream piezometers may periodically get submerged due to large increases in stage allowing surface water to enter vent holes or caps causing erroneous data (Figure 1a). Installation of piezometer transects across some river systems may be impractical due to the depth of water or risk of tampering in urban environments. Piezometers can be capped at the sediment-water interface to avoid visibility and damage from flood events, but pressure data would be compromised. Thermocouple wires connected to data loggers were once a common practice [e.g., *Stonestrom and Constantz*, 2003], but require expensive external data-loggers and battery power be placed in large insulated enclosures near the measuring point either above the water surface (Figure 1b) or on the bank. Thermocouple arrays are difficult to install and can suffer damage to electronics and wires during high flow events. Shifts in the thermocouple wires during monitoring can also cause temperature anomalies. Recent advances in FO-DTS have resulted in the ability to measure temperatures at fine vertical resolutions in both the surface and subsurface sediments [*Vogt et al*., 2010; *Briggs et al*., 2013]. However, the use of FO-DTS has significant power supply requirements and it has not reached the point where this technology can be used autonomously (Figure 1c). Similarly, in-stream deployments may be susceptible to debris on the fiber optic cable led back to the DTS control unit or between sites if more than one location is being monitored [*Vogt et al*., 2010]. The costs associated with FO-DTS deployments are significant (over $35,000 USD) and may be prohibitive for most studies. FO-DTS is not easy to install, requires extensive data processing, and system output requires frequent or continuous calibration to minimize measurement drift and uncertainty [*Rose et al*., 2013]. Ideally, the use of accurate low-cost autonomous probes that can be deployed for long periods of time and in a variety of flow regimes are needed for environmental studies of riverine and hyporheic systems (Figure 1d).

The goal of this methods paper is to present a new subsurface temperature profiling probe (referred to herein as TROD, from temperature rod) that was designed for monitoring temperature at 6 discrete temperature depths along a ¾ inch diameter 1 m long sealed PVC pipe (Figure 1d). The temperature sensors are secured inside a water-proof enclosure to avoid moisture damage to electronics. The TROD is designed to be installed into stream sediment for measuring temperature in both sediment and/or surface water. This temperature probe is advantageous over other methods, because of (1) the low-profile design allows for continuous monitoring in harsh environments that are susceptible high-discharge events when damage from sheer or rapidly moving sediment or debris is likely, (2) does not require removal or disruption for data retrieval, (3) uses low-cost existing self-logging temperature sensors that are easily replaced and allows prolonged use the components, and (4) developed user friendly software efficient in programming and retrieval of temperature data. A version of the TROD has been constructed for use without the communication cable for situations where visibility may expose the instrumentation to vandalism (i.e., urban streams) or where burying the cable may be impractical (i.e., dynamic lake shore environments). The sensor spacing interval and depth can be chosen to optimize observation density for a variety of applications. We present a description of the materials, test the accuracy before and after calibration, and demonstrate the use for estimation of seepage losses in an unlined agricultural canal.

2. Materials, Software, and Testing

The TROD can generally be described as a set of six self-logging iButtons placed at fixed locations inside a ¾ inch diameter schedule 80 PVC pipe with a solid PVC drive point and a 2 inch diameter head that is sealed to resist the ingress of water (Figure 2). The iButtons are connected in series to a heavy duty submersible communication cable (16 AWG Super VU-TRON Supreme Type S00W) that allows for downloading the temperature data via USB port. Without the cable, data can also be retrieved the head of the probe while the probe is submerged (Figure 2b). We tested communication cable lengths 1.5–7.6 m (5–25 ft) with the assumption that data retrieval could occur at a variable distance to TROD. Without the communication cable, connection is made using a commercially available iButton reader (*Connection Clamp*, Alpha Mach, Inc). The iButtons are positioned on a printed circuit board (PCB) just beneath the head at a depths of 0.0, 0.1, 0.2, 0.5, 0.75, and 1.0 m. However, the sensor spacing and location are flexible and can be customized to any depth at an interval spacing of 0.05 m. They are positioned on the PCB to minimize the distance between the inner wall of the pipe and top of the iButton sensor where the temperature is measured.

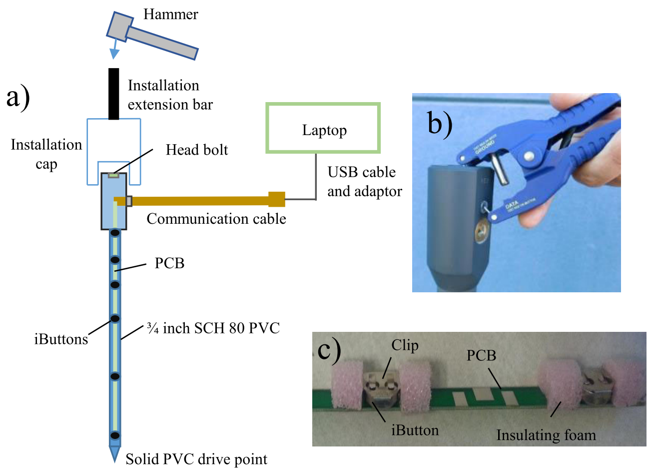

Figure 2. Schematic of the (a) subsurface temperature profiling probe (TROD) and (b) the iButton reader (communication clamp, Alpha Mach, Inc) used to program the sensors and retrieve data without the cable, and (c) printed circuit board (PCB) with iButtons connected in series. The extension cap and bar are used to assist installation of the probe by direct hammering when the top of the instrument is under water. Either the communication cable in Figure 2a or the iButton reader shown in Figure 2b, can be used to program the TROD and retrieve data while the probe is submerged. iButtons are replaced by removing the head bolt, extracting the PCB, and fastening new ones into the retaining clips.

Above and below each sensor, nonconductive foam cylinders prevent thermal conduction inside the probe and improve the precision of the temperature measurements (Figure 2c).

The TROD was designed with the recognition that the low-cost iButtons will likely need replacing every 2–4 years, depending on use. In this probe, the 6 iButtons can be easily replaced by removing the head bolt and pulling out the PCB assembly, and sliding them out of the clip holders (Figures 2a and 2c). The PCB clip holders allow the sensors to be snapped into place for easy removal and replacement. The TROD design provides: (1) the ability to replace the sensors without needing any rewiring or soldering; (2) the ability to add new sensors at other depth intervals; and (3) eliminates the possibility of losing connection during field installation because once the sensors are snapped into place, they do not become dislodged. The iButtons remain at fixed locations and will not shift or move during installation or monitoring. Once reassembled, it is necessary to reapply sealant to the threads on the head bolt to prevent water from entering the TROD.

A software application, *WeeButton* (www.alphamach.com/software-downloads) was developed specifically for the TROD to simultaneously program and download temperature data from all iButtons and export data to an Excel spreadsheet. Functionality of software also includes storage of iButton unique serial numbers, calibration corrections, and TROD identification. Therefore, upon connecting to each TROD, the *WeeButton* software searches for each iButtons unique serial numbers stored within the database and automatically applies the sensor specific calibration corrections. During each download, the software recognizes the TROD number and position of the iButtons and formats the output into a table with date, time, and respective calibrated temperatures for each depth. The program can customize settings for each depth on the TROD for variable time monitoring. Calibration corrections can also be provided by Alpha Mach, Inc for a small fee.

The accuracy and precision of iButtons are model specific and are available from the manufacturer; however, the thermal properties of the PVC will have an effect on the response time. The response time was measured by submerging the entire TROD, initially at room temperature, into a section of 4-inch diameter PVC pipe in equilibrium with a temperature controlled water bath at 5.0°C. The thermal time constant (τ) or elapsed time for the temperature to change by 63.2 percent between room temperature and bath temperature, was calculated by a linear regression of the time rate of temperature change against time [*Whiteman et al.*, 2000]. Six polyurethane sealed iButtons were attached to the outside of the TROD adjacent to internal sensors for comparison of the response and thermal time constant.

The probe also was tested for accuracy by circulating water around the TROD in the same pipe used to test for response time. The accuracy was evaluated by computing the difference in temperatures measured by the TROD sensor and a NIST certified traceable digital thermometer (Cat 4000, Control Company) with resolution (0.0001°C) and accuracy (0.05°C). Individual iButtons in the TROD were evaluated at target bath temperatures of 5, 10, 15 and 20°C. The temperatures used in the calibration spans the published range of DS1921Z Type Z iButton (−5°C to 26°C) and is intended to improve the accuracy from ± 1.0°C to 0.1°C [*Johnson et al.*, 2005; *Hester et al.*, 2009; *Naranjo et al.*, 2012]. The temperature residual, or difference between the NIST thermometer and each sensor on the probe, was evaluated pre and postcalibration.

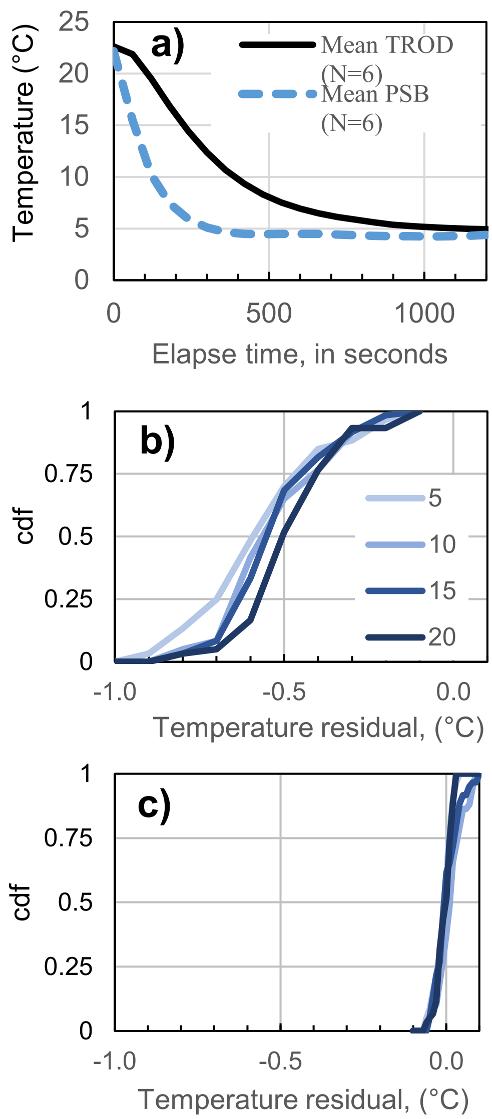

Figure 3. (a) Temperature response of a TROD compared to six polyurethane sealed iButtons (PSB) following full submersion from room temperature of 20.1 to bath temperature of 5.0°C. Cumulative distribution functions (cdf) for (b) precalibration and (c) postcalibration temperature residuals (probe - NIST) at target bath temperatures of 5.0, 10.0, 15.0 and 20.0°C for 10 temperature probes (N = 60). The range of the median residual temperature for precalibration is −0.51 to −0.59°C and for postcalibration 0.017–0.012°C, respectively.

3. Results and Discussion

3.1. Laboratory Comparison Results

Temperatures measured with the TROD and the polyurethane sealed iButtons (PSBs) following submersion in the water bath and results of pre and postcalibration residuals are shown in Figure 3. The range of τ for each of the 6 individual sensors in the TROD were 194 - 268 s with a median of 217 ± 28 s. The six PSBs directly exposed to the bath water adjacent to the sensors inside the rod responded more rapidly with a median τ of 77 ± 13 s (Figure 3a). The median elapsed time for the all sensors in the TROD to reach 5.0°C was 1000 s. The PSBs reached the temperature of 5.0°C in 360 s. The τ of the TROD is increased by the insulating properties of the PVC and air space inside the TROD. However, the TROD compares well with autonomous temperature sensors by HOBO Pro V2 (τ = 122 s) [see *Young et al.*, 2014] and Tidbit v2 (τ = 300 s; Onset Stowaway) not housed in piezometers or pipes. It is likely these temperature sensors would have a longer τ if tested inside a piezometer or pipe.

The improvements in accuracy and sensor resolution through individualized calibration is an important consideration for resolving differences in measurements between sensors. The residual error between the NIST thermometer and 10 TRODs (n=60) for the four target temperatures for are shown in Figures 3b and 3c. A negative bias was observed for precalibrated iButtons for all target bath temperatures with a range in the median residual from −0.51 to −0.59°C (Figure 3b). The residuals for all sensors were under the manufacturer's specifications of ± 1.0°C. Improvements to the accuracy were attained by applying a linear regression to each of the six iButtons within the TROD. The reduction in residuals for each of the target bath temperatures was less than 0.1°C with a range in median residual temperature of −0.017 to 0.012°C (Figure 3c). These results show that the water bath calibration significantly improves the accuracy of the iButtons when performed for each sensor separately.

3.2. Example Application

The TROD was installed in an unlined agricultural canal in the Walker River Basin, north-central Nevada for use in seepage estimation during an entire irrigation season (Figure 4). The TROD was installed in conditions with and without flowing water into sediment consisting of silty loam to coarse sand texture with cobbles. A piloted hole was used to install the TROD in both dry and flowing conditions to prevent breakage and to ensure the completion depth of 1m could be attained (Figures 4a and 4b). In flowing conditions, the extension bar connected to the installation cap assisted installation above and below the surface water. Direct hammering with a sledge hammer or a fence post driver was used until the top iButton temperature sensor was located at the sediment-water interface (Figure 4c). Canal stage was recorded using a pressure transducer (Diver; Slumberger) compensated for barometric fluctuations. Temperature data were inspected following the installation of the TROD for signs of preferential flow downward along the pipe.

Figure 4. (a) Demonstration of the TROD being installed in an irrigation canal with a pilot hole, in (b) flowing surface water and (c) a completed transect with two probes installed. The pilot hole was filled with a bentonite slurry prior to inserting the TROD to prevent vertical preferential flow down the pipe. The low-profile design allows monitoring during the entire irrigation season while avoiding debris entanglement that may disrupt data collection. Photos taken from agricultural canals in the Walker River Basin, north-central Nevada. The 25 ft communication cables in these examples were buried in the sediments and data were retrieved from the bank of the canal.

Temperature and stage data were recorded at 30 min intervals over two irrigation seasons. Data collected during the 2012 irrigation season are shown in Figure 5. Continuous temperature data show short term (diel) and longer term (weekly to annual) thermal patterns as well as a gradients reversal on 29 August 2012 from warmer near surface ($<$ 0.5m) temperatures to cooler temperatures. Using traditional methods to collect temperature data over the entire irrigation season would have required extensive efforts to remove debris and frequent removal of sensors for retrieving data. This continuous data will be used for developing time-varying seepage estimates using numerical modeling. The use of the low-profile TRODs allowed unobstructed monitoring completely unaffected by irrigation activities that will improve annual estimates of seepage loss.

3.3. Limitations

Sample intervals defined for each iButton in the TROD can be defined as low as 1 min with frequent data retrieval to avoid maximizing the memory storage of the sensors. For example, at hourly sample intervals, the TROD with Type Z iButtons can collect data for 85 days or 2048 observations before storage has been depleted. It is also possible to use more costly ($70 USD) higher precision ($\pm$0.5°C) and resolution (0.0625°C) iButtons (DS1922L; Type L) that can store twice as many observations.

The response time of the TROD is longer than temperature sensors directly in contact with water. Evaluations of τ were not tested for other temperature loggers (e.g., HOBO, Tidbit, iBCods) placed inside pipes for a cross-sensor comparison. However, the τ for the TROD is 84 s less than what is reported for Tidbit v2 and 96 s longer than the HOBO Pro V2, both tested in direct contact with water. It is expected that the Tidbit and HOBO temperature sensors would have a longer τ if tested inside a pipe or piezometer. A delay in the response time can also be relevant to temperature sensor deployments in piezometers. As reported in *Cardenas* [2010] the delayed response in and through piezometers of 1–3 inch diameter PVC may cause uncertainty in the estimation of flux caused by the thermal skin effects on the timing of the thermal signal. The response time and accuracy of the TROD is suitable for most applications in hydrology but careful consideration of the increased response time is warranted for each particular application.

Installation of the TROD into cobble bed environments may prove difficult as with other methods of monitoring subsurface temperatures. The TROD is a robust probe that can be installed in a variety of substrates but to prevent breakage a pilot hole should be used to loosen sediments and ensure that the desired depth can be obtained. Sealing with bentonite slurry may also be necessary to avoid preferential flow vertically along the TROD.

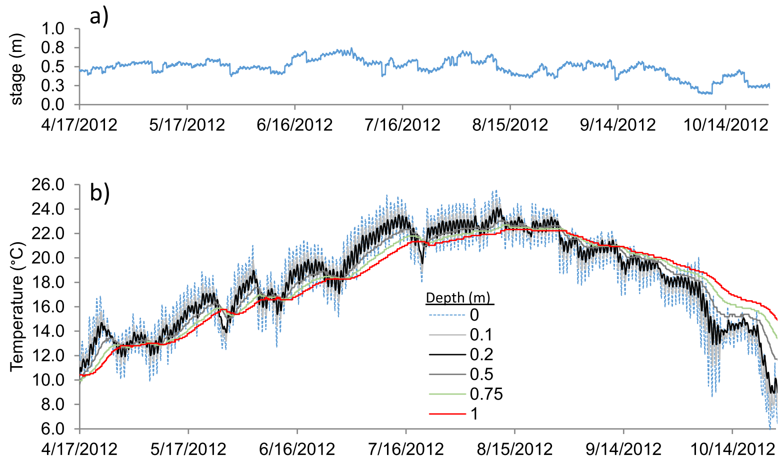

Figure 5. (a) Surface water stage and (b) sediment temperatures measured with a TROD during the 2012 irrigation season located in an unlined irrigation canal in the Walker River Basin, north-central Nevada. Temperature observations are in meters below the sediment-water interface and stage recorded with a pressure transducer compensated for barometric influences.

3.4. Advantages

The TROD largely overcomes the limitations of other equipment used for measuring environmental temperatures, especially in stream environments. The main advantages are (1) the low-profile design allows for continuous monitoring in harsh environments that are susceptible to floating debris, high velocities or stage, (2) does not require removal or disruption for data retrieval, (3) uses low-cost existing self-logging temperature sensors that are easily replaced and allows prolonged use of the bulk of materials, and (4) developed software specifically for the probe is easy to use, efficient in programing and downloading data.

The TROD also is suitable for long-term deployments for assessing time varying seepage in ephemeral channels, and hyporheic zones, evaluating the effects of channel scour and deposition [*Tonina et al*., 2014], and snow melt and infiltration processes on the landscape [*Lundquist and Lott*, 2008]. The flexibility in specifying a high temporal resolution and placing iButtons at small interval spacing (0.05 m) are important considerations for some environmental conditions, such as upwelling zones in river systems [*Briggs et al*., 2014]. The low-profile design allows the monitoring of sediments in deeper areas of surface water bodies, such as pools without the instrument protruding above the sediment-water interface. This is a significant advancement in the investigations of large river systems where debris and flood events may cause instrument damage. With the development of new software, programming all the iButtons in the TROD and retrieving data is time-efficient compared to downloading each separately. Following installation, temperature data can be inspected for anomalous readings created by preferential flow vertically down the sides of the TROD. The ability to download data periodically while the probe remains in place is advantageous in limiting the disturbance of continuous time series records and identifying if the installation was compromised. Removing and replacing inexpensive temperature sensors (iButtons) allows repeated use of the probe. The ability to measure sediment temperatures in a variety of flow and environmental conditions over longer periods of time opens up opportunities for new research in hydrological and soil sciences. This is especially true for the collection of spatially variable data needed for multidimensional models.

4. Summary

A newly developed subsurface temperature profiling probe (TROD) is an innovative and useful tool for examining thermal behavior beneath the sediment-water interface. This temperature probe opens up a range of possible new applications in hydrology, soil science, agriculture, and stream ecology. The main advantages as described are the design, materials and the developed software. The materials and sensors used in its design are accessible and affordable. The ability to instrument the TROD in a variety of environments will allow temperature monitoring in areas that were overlooked due to the practical limitations of installation or susceptibility to damage.

Acknowledgments
Data generated for this manuscript can be requested from the corresponding author. The design of this probe was a collaborative effort between the U.S. Geological Survey and Alpha Mach, Inc. This technology has patent pending components. The early conceptualization of this probe is based on input and ideas expressed by many people including Richard Niswonger and David Stonestrom of the U.S. Geological Survey, James Brock and Greg Pohll of the Desert Research Institute. We would like to thank Richard Niswonger and David Stonestrom for suggestions on improving the manuscript. We appreciate the work of Steve Clark for development of Matlab code for data processing and Dave Smith of the U.S. Geological Survey for field collection. We also thank the AE and the anonymous reviewers who provided insightful comments that improved this manuscript. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

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